

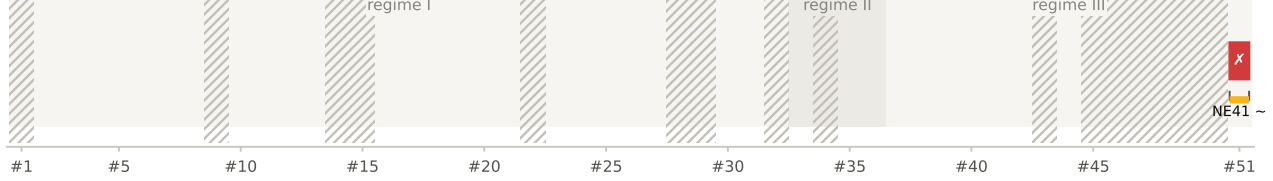
H130: #51 agents are Ornstein–Uhlenbeck particles in private wells: reads kick them and the kick decays at the well’s rate

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. In #51 a read pulls the next statement toward the sender (J_K 0.044 [0.038, 0.050], 11/12 units, both models), but the pull fades about 15× faster (≈ 7 calls) than the agent’s own content memory (γ_{auto} 0.0094/call, homogeneous over 12 units).

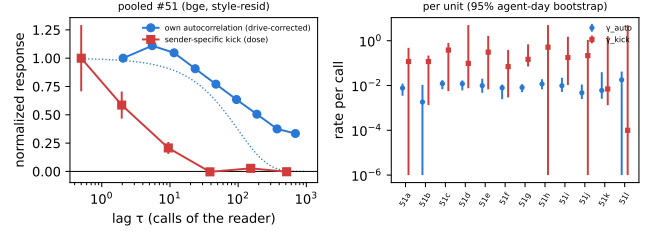
level: descriptive · pipeline checks 86% · updated 2026-10-04



Goal periods tested (1 failed, 1 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending.
Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

HH371 proposed the simplest kinetic version of H98’s random field: each #51 agent is an Ornstein–Uhlenbeck particle in its own well, every read is a small kick toward the sender, and the kick decays at the well’s own rate. H130 measures both rates on the per-call clock in all 12 non-holdout #51 units (45 days, 40,958 statements, 930k reads), with two embedding models. A read does kick: the read-minus-in-flight jump is $J_K = 0.044$ [0.038, 0.050], positive in every unit with ≥ 3 days. The well relaxes at $\gamma_{\text{auto}} = 0.0094$ per call (≈ 100 calls, 1 h), the same in every unit. The kick fades $\approx 15\times$ faster (ρ_γ 14.8, 90% CI [7.2, 30.4]), so the one-rate kill fires. The holdout is not run.



Observables. Left: pooled #51 (bge). Own content autocorrelation, corrected for the common drive (blue), against the sender-specific kick per read from the dose regression (red), both normalized, against lag in the reader’s calls. The dotted line is $e^{-\gamma_{\text{auto}}\tau}$. The kick is gone after ≈ 10 calls; the memory lasts ≈ 100 . Right: per-unit rates with 95% agent-day bootstrap CIs.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Vector OU dynamics on the call clock (model 11) with a measured response tested against fluctuations (model 02). x_i is agent i ’s 32-d whitened, style-residualized content minus its well centre h_i (leave-day-out mean). γ is the well stiffness per call, κ the kick per read toward the sender’s message z_m , ξ the agent’s own noise, D_r a room drive. A linear system has one rate for response and memory; a kick held only in the context window would decay faster.

$$x_i(n+1) = (1 - \gamma) x_i(n) + \kappa \sum_{m \text{ read at } n} (z_m - h_i) + \xi_i(n)$$

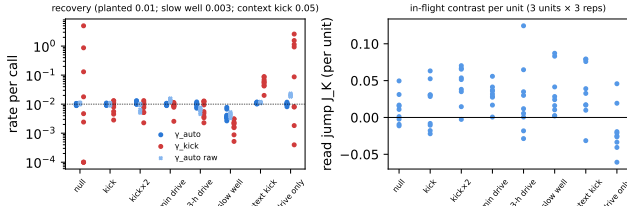
$$C(\tau) = \langle x_B \cdot x_{B'} \rangle - C_\times(|t_{B'} - t_B|) \rightarrow \gamma_{\text{auto}}$$

$$y_{Bj} = x_B \cdot \widehat{h_j} - h_i = \sum_b \beta_b R_b^{(j)} + \dots \rightarrow \gamma_{\text{kick}}, \quad \rho_\gamma = \gamma_{\text{kick}} / \gamma_{\text{auto}}$$

H130-ou-private-wells-51 · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H130-ou-private-wells-51/README.md · ratings: H130 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1/K2 read kick > in-flight	0.044 [0.038, 0.050]; 7/7 units ≥ 3	supported
P2/K1 $\rho_\gamma \in [0.5, 2]$	14.8 [7.2, 30.4]; gte 20.2	failed (kill)
P3 co-movement rises with reads	β_R 0.018 [0.012, 0.024]	supported
P4 pair decay at γ	$\gamma_\times / \gamma_{\text{auto}} \approx 6-8$	failed
P5 one well rate, 30-600 calls	I^2 0; $1/\gamma \approx 106$ calls	supported
P6 call clock fits better	5/12 units	not supported
N1 rival kick ratio $\in [0.5, 2]$	2.4 [-0.7, 5.3]	inconclusive
N2 NE41: memory, R_C 1.03 [0.95, 1.10]; R_K undefined	kick survive erasure	mixed
N3 joiners relax at γ	flat at 0.39 of own well on day 1	failed



Second figure. Synthetic validation on the real #51 skeleton (real statements, calls, reads, resets; 3 units $\times$ 3 replicates). The drive-corrected γ_{auto} recovers the planted 0.01 under 5-min to 3-h room drives; the raw estimator does not. The sender-specific γ_{kick} recovers 0.01 and separates a context-held kick (0.05). The registered message-direction profile read the drive's rate and was replaced before real data (A1).

GOAL PERIODS IN DETAIL

G51 $\times$ failed
NE41 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The estimators for both rates were changed after the synthetic and before real data (A1). γ_{kick} rests on the first 2–3 lag bins, so its CI spans $\times 4$, but the lower 90% bound of ρ_γ is 7. In one-room #51 every agent reads every message, so beyond the first read-out the kick is identified only through sender-specific doses. The erasure test of the kick is undefined (the kick has decayed by lag 4). One private-role period; no transfer test.

ROUND 2

Resolve the kick kernel at 1-call steps and split replies from plain reads; fit joiners' well formation over days; run the read jump and the two rates on #38–#44; run the frozen confirm on the #51 tail after sign-off.

- **Essence.** what clock the swarm's influence runs on, separate from what clock its memory runs on.
- **H130-R1** Fast-kick kernel.
- **H130-R2** Kick vs reply.
- **H130-R3** Well formation.
- **H130-R4** Transfer.